EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Chemical Properties of Group 13 Elements (Reactivity towards Water, Air, Acids, Alkalies, Metals, Non-Metals and Halogens)

1. $2E + N_2 \longrightarrow 2EN$ (very hard substance)

$EN + H_2O \longrightarrow$ Acid + pungent smelling gas

Acid is :

(A) HNO_3 (B) H_3BO_3 (C) HNO_2 (D) can be A & B

CPB179

2. $\text{Boron} \xrightarrow{Z} x \xrightarrow{LiH} y(g) + LiBF_4$, with $x \xrightarrow{W} \text{Boron}$

true statement is :

(A) z is least reactive non-metal (B) w is cooling

(C) y is electron deficient (D) All of these

CPB180

3. An element E = {B, C, Si, Ge}, predict E on the basis of given conditions:

I : Powdered E reacts with O_2 to form an oxide.

II : Oxide formed in I reacts with NaOH

III : E reacts with steam on red heat forming two gaseous products, which can be used as a fuel.

E can be :

(A) B (B) C (C) Si (D) Ge

CPB193

4. Boron reacts with air to give:

(A) B_2O_3 (B) BN (C) Both (A) & (B) (D) None

CPB113

5. Aluminium is not acted upon by pure water as :

(A) Impurities in water are essential for the reaction to occur

(B) It is light metal

(C) It is protected by a film of a aluminium oxide

(D) It is not a reactive metal

CPB114

6. The final product obtained when boric acid is heated to red heat is –

(A) Metaboric acid (B) Tetra boric acid

(C) Boric anhydride (D) Pyroboric acid

CPB116

Pankers Academy JEE

For More Join: @IITJEE_Advanced

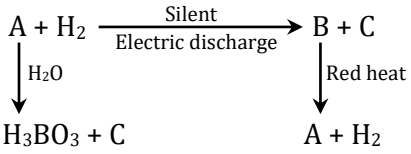
Preparation of Boron Some Important compounds of Group-13 and their physical and chemical properties (Borax, Diborane, Orthoboric Acid, Alums)

7. Aqueous solution containing 1 mol of borax reacts with 2 mol of acids. This is because of :
(A) formation of 2 mol of $B(OH)_3$ only
(B) formation of 2 mol of $[B(OH)_4]^-$ only
(C) formation of 1 mol each of $B(OH)_3$ and $[B(OH)_4]^-$
(D) formation of 2 mol each of $[B(OH)_4]^-$ and $B(OH)_3$, of which only $[B(OH)_4]^-$ reacts with acid

CPB178

8. $B_2H_6 + NH_3 \longrightarrow$ Addition compound (X) $\xrightarrow{450\ K}$ Y + Z (g)
In the above sequence Y and Z are respectively :
(A) borazone, H_2 (B) boron, H_2
(C) boron nitride, H_2 (D) borazine and hydrogen

CPB181

9.


Identify correct statement.
(A) C is a weak lewis acid (B) B is a weak lewis base
(C) C is a strong acid (D) D reacts with NaOH to produce C

CPB183

10. $4BCl_3 + 3LiAlH_4 \longrightarrow A + 3AlCl_3 + 3\ LiCl$
When A reacts with NaOH it produces a colourless combustible gas and another compound 'B'. Select incorrect statement about 'B'.
(A) It aqueous solution turns red litmus blue
(B) It shows anionic hydrolysis
(C) It shows cationic hydrolysis
(D) It can also produce by reaction of boron with NaOH

CPB184

11. Diborane readily reacts with water forming –
(A) Diborate and O_2 (B) Boric acid and O_2
(C) Boric acid and H_2 (D) Diborate and H_2

CPB130

12. Which one of the following compounds gives methane on treatment with water :-
(A) Al_4C_3 (B) CaC_2 (C) Mg_2C_3 (D) SiC

CPB166

13. A mixture of Boric acid with Ehtyl alcohol burns with green edged flame due to the formation of –
(A) Ethyl borax (B) Ethyl borate (C) Methyl borax (D) Methyl borate

CPB118

14. Borax $Na_2B_4O_7.10H_2O$ is actually:-
(A) $Na_2[B_4O_5(OH)_4].8H_2O$ (B) $Na_2[B_4O_4(OH)_6].7H_2O$
(C) $Na_2[B_4O_3(OH)_8].6H_2O$ (D) $Na_2[B_4O_2(OH)_{10}].5H_2O$

CPB120

For More Join: @IITJEE_Advanced

Uses of Boron and Aluminium and their compounds

15. Borazine and benzene show striking similarities in their properties. This led to a labeling of borazine as 'inorganic benzene'. However, in contrast to benzene, borazine readily undergoes addition reactions. The appropriate reason for this difference is :

(A) Alternate arrangement of B and N atoms in the hexagonal ring.

(B) Due to the difference in electronegativity between boron and nitrogen, the cloud (electron density) is more localised on the nitrogen atoms.

(C) Aromatic π-clouds of electron density are delocalised over all of the ring atoms.

(D) Net dipole moment of borazine molecule is non-zero.

CPB182

16. Bauxite and cryolite are the minerals of which elements?

(A) B (B) Al (C) Ga (D) In

CPB112

17. Which of the following does not occur as solid metal at near room temperature?

(A) B (B) Al (C) Ga (D) In

CPB110

18. Which one of the following mixed sulphates is not an alum?

(A) $K_2SO_4.Al_2(SO_4)_3.\ 24H_2O$ (B) $K_2SO_4.Cr_2(SO_4)_3.\ 24H_2O$

(C) $(NH_4)_2SO_4.Fe_2(SO_4)_3.\ 24H_2O$ (D) $CuSO_4.Al_2(SO_4)_3.\ 24H_2O$

CPB132

Rankers Academy JEE

Chemical Properties of Group 14 Elements (Reactivity towards Oxygen, Water, Acids, Alkalies, Hydrogen, Halogen)

19. Which of the following order is correct?

(A) $SiCl_4 > SiF_4 > SiBr_4 > SiI_4$ (Lewis acid strength)

(B) $H_3BO_3 > Na[B(OH)_4]$ ($d_{B\text{-}O}$ bond strength)

(C) $CHCl_3 < CHF_3$ (Acidic nature)

(D) $BeF_2 > BeCl_2 > BeBr_2 > BeI_2$ (Lewis acid nature)

CPB133

Allotropes of Carbon

20. Purest allotrope of carbon is :

(A) Diamond (B) Graphite (C) Fullerene (D) Coal

CPB147

21. Given the order of density : Diamond > Graphite > Fullerene C_{60} ; choose the correct order for C–C bond length

(Consider larger bond length if there are two different bond lengths)

(A) Diamond < graphite < fullerene (C_{60}) (B) Diamond > graphite > fullerene (C_{60})

(C) Diamond > fullerene (C_{60}) > graphite (D) Diamond < fullerene (C_{60}) < graphite

CPB189

For More Join: @IITJEE_Advanced

22. Which is not the property of diamond?

(A) It is insoluble in all solvents

(B) It is oxidised with a mixture of $K_2Cr_2O_7$ and H_2SO_4 at 200ºC

(C) Being hardest, it is used as an abrasive for sharpening hard tools.

(D) $\Delta_f H^{\ominus}$ value of diamond is 1.90 kJ mol^{-1}.

CPB194

Some important Compounds of Carbon: CO and CO_2

23. Select the correct option regarding the bond strength of $C^{14}O^{16}$, $C^{12}O^{16}$, $C^{14}O^{18}$.

(A) $C^{14}O^{18} > C^{14}O^{16} > C^{12}O^{16}$ (B) $C^{14}O^{18} < C^{14}O^{16} < C^{12}O^{16}$

(C) $C^{14}O^{18} > C^{14}O^{16} < C^{12}O^{16}$ (D) $C^{14}O^{18} < C^{14}O^{16} > C^{12}O^{16}$

CPB192

Silicon and its Compounds SiO_2

24. CCl_4 is inert towards hydrolysis but $SiCl_4$ is readily hydrolysed because

(A) carbon cannot expand its octet but silicon can expand its octet

(B) ionisation potential of carbon is higher than silicon

(C) carbon forms double and triple bonds

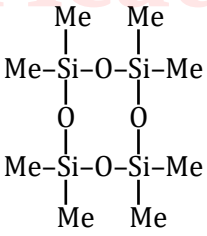
(D) electronegativity of carbon is higher than that of silicon

CPB190

Silicones

Rankers Academy JEE

25. Given type of silicones are called [P]



[P] is prepared by controlled hydrolysis of [Q]. [P] & [Q] are respectively.

(A) Linear silicone, CH_3SiCl_3 (B) branched silicone, $(CH_3)_3SiCl$

(C) Cyclic silicone, $(CH_3)_2SiCl_2$ (D) Cyclic silicone, CH_3SiCl_3

CPB188

Silicates

26. The structural unit present in pyrosillicates is :

(A) $Si_3O_9^{6-}$ (B) SiO_4^{4-} (C) $Si_2O_7^{6-}$ (D) $(Si_2O_5^{2-})_n$

CPB185

Carbides

27. The number of types of bonds between two carbon atoms in calcium carbide is :-

(A) One sigma, two pi (B) One sigma, one pi

(C) Two sigma, one pi (D) Two sigma, two pi

CPB167

For More Join: @IITJEE_Advanced

28. The ionic carbide is :-

(A) ZnC (B) TiC (C) SiC (D) CaC_2

CPB165

Compounds of Lead

29. Among the following, Amphoteric oxides are :

(A) SiO, SiO_2, GeO_2 (B) SnO, SnO_2, PbO_2 (C) CO, SiO_2, CO_2 (D) None of these

CPB144

30. E represents an element belonging to carbon family.

$E + 2X_2 \longrightarrow EX_4$ (X = F, Cl, Br, I)

(A) Stability of EX_4 decreases down the 14th group

(B) PbI_4 does not exist

(C) Ge & Pb forms EX_2 as well.

(D) All are correct

CPB186

31. When heating white lead then find out released gas (A) and (B)

$2PbCO_3.Pb(OH)_2 \xrightarrow{\Delta} Pb_3O_4 + (A) + (B) + H_2O$

(white lead)

(A) CO, O_2 (B) CO_2 , O_2 (C) CO_2, CH_4 (D) CO, CO_2

CPB187

32. PbF_4, $PbCl_4$ exists but $PbBr_4$ and PbI_4 do not exist because of :

Raiker's Academy JEE

(A) Large size of Bi^- and I^-

(B) Strong oxidising character of Pb^{+4}

(C) Strong reducing character of Pb^{+4}

(D) Low electro negativity of Br^- and I^-

CPB191

33. Which compound is known as Litharge?

(A) PbO_2 (B) $Pb(NO_3)_2$ (C) PbO (D) $PbSO_4$

CPB170

34. Which element known as the softest heavy metal ?

(A) Copper (B) Iron (C) Lead (D) Zinc

CPB171

35. Sulphide ore of Lead is known as:

(A) Pyrite (B) Galena (C) Cinnabar (D) Copper glance

CPB172

Group 15 Elements (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

36. The thermal stability of the hydrides of group 15 follows the order :

(A) $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$ (B) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(C) $PH_3 > NH_3 > AsH_3 > SbH_3 < BiH_3$ (D) $AsH_3 < PH_3 > SbH_3 > BiH_3 > NH_3$

CPE289

For More Join: @IITJEE_Advanced

PCl_5 Preparation and Properties

37. PCl_5 on reaction with alcohol gives :
(A) Alkyl halide (B) Alkane (C) Dihalides (D) Alkene

CPE145

Group 16 Elements (Occurance, Electronic configuration, Atomic Radii and Ionic Radii. Ionisation Enthalpy, Electrongain Enthalpy, Electronegativity, Physical Properties)

38. The correct order of electron affinity is :
(A) O > S > Se > Te > Po (B) Po > Te > Se > S > O
(C) S > O > Se > Te > Po (D) S > Se > Te > Po > O

CPE163

39. The group 16 element with highest ionisation energy:
(A) S (B) O (C) Se (D) Po

CPE164

40. H_2O is a liquid while H_2S is a gas. The reason is:
(A) H_2O has higher van der waal forces (B) Compact nature of H_2O
(C) Hydrogen bonding in H_2O (D) Electronegativity difference.

CPE166

Dioxygen Preparation and Properties

41. O_2 molecule is :
(A) Diamagnetic (B) Polar
(C) Paramagnetic (D) Odd electron molecule

Rankers Academy JEE

CPE167

42. Oxygen is industrially prepared by:
(A) Decomposition of H_2O_2
(B) Electrolysis of water
(C) Decomposition of permanganates, chlorates and nitrates
(D) Fractional distillation of air

CPE168

Ozone Preparation and Properties

43. Pick out the statement which is incorrect:
(A) Oxygen is paramagnetic in all the three states of matter
(B) Ozone is diamagnetic
(C) Ozone is a linear molecule
(D) The O–O bonds in Ozone have partial double bond character

CPE175

44. Which one of the following property is not correct for Ozone?
(A) It oxidises lead sulphide
(B) It oxidises potassium iodide
(C) It oxidises mercury
(D) It cannot act as bleaching agent in dry state.

CPE174

For More Join: @IITJEE_Advanced

45. Match the column-I and column-II :

Column – I	**Column – II**
(a) Ozone gas	(p) Violet – black
(b) Ozone Liquid	(q) Pale blue
(c) Ozone solid	(r) Dark – blue

(A) a → q ; b → r ; c → p (B) a → r ; b → p ; c → q

(C) a → p ; b → q ; c → r (D) a → r ; b → q ; c → p

CPE176

Allotropes of Sulphur

46. The transition temperature of α – Sulphur to β-Sulphur is :

(A) 393 K (B) 369 °C (C) 369 K (D) 385.8 K

CPE189

47. The S_6 molecule has :

(A) Ring form (B) Chair form (C) Hexagonal form (D) Crown form

CPE191

SO_2 Preparation and Properties

48. $2SO_2 + O_2 \xrightarrow{'X'} 2SO_3$, the catalyst 'X' is :

(A) $CuCl_2$ (B) Ni (C) Pt (D) V_2O_5

Rankers Academy JEE

CPE169

49. By which of the following, SO_2 is formed?

(A) Reaction of dilute H_2SO_4 with Zn (B) Heating of $Fe_2(SO_4)_3$

(C) Reaction of concentrated H_2SO_4 with Cu (D) Heating of $CuSO_4$

CPE193

50. Roasting of sulphides gives the gas X as a by-product. This is colourless gas with choking smell of burnt sulphur and causes great damage to respiratory organs as a result of acid rain. Its aqueous solution is acidic, acts as reducing agent and its acid has never been isolated. The gas X is :

(A) SO_3 (B) H_2S (C) SO_2 (D) CO_2

CPE194

51. On reaction of SO_2 with acidic $KMnO_4$ solution, which is correctly observed?

(A) Colour of $KMnO_4$ is disappeared (B) SO_2 is oxidised to SO_3

(C) $MnO_4^{\ominus}$ is reduced to MnO_2 (D) All of the above

CPE195

Uses of Group - 16 Elements and Compounds

52. Salt cake is:

(A) Na_2SO_4 (B) NaCl (C) $NaHSO_4$ (D) $NaHSO_4$ + HCl

CPE242

For More Join: @IITJEE_Advanced

Anomalous behaviour of Fluorine; Chemical properties of Group 17 Elements (Reactivity towards Hydrogen, Oxygen, Metals and other Halogens)

53. Match the Column-I and Column-II :

Column – I	**Column – II**
(a) F_2	(p) Green
(b) Cl_2	(q) Violet
(c) Br_2	(r) Yellow
(d) I_2	(s) Red

(A) $a \rightarrow p; b \rightarrow r; c \rightarrow q; d \rightarrow s$ (B) $a \rightarrow r; b \rightarrow p; c \rightarrow s; d \rightarrow q$

(C) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow s$ (D) $a \rightarrow r; b \rightarrow s; c \rightarrow p; d \rightarrow q$

CPE215

54. Which of the following exist (s) in gaseous state?

(A) F_2 (B) F_2 and Cl_2 (C) Cl_2 and Br_2 (D) F_2, Cl_2 and I_2

CPE216

55. Which is a super Halogen?

(A) F (B) Cl (C) Br (D) I

CPE219

56. Which is a powerful oxidising agent?

(A) F_2O_2 (B) Cl_2O_6 (C) BrO_3 (D) I_2O_5

CPE220

Rankers Academy JEE

Chlorine: Preparations and Properties

57. Chlorine is manufactured by :

(A) Birkeland and Eyde process (B) Deacon process

(C) Bosch process (D) Solvey process

CPE222

58. Chlorine gas is not used in :

(A) Bleaching agent (B) Extraction of gold

(C) Refining of sugar (D) Sterilising drinking water

CPE224

59. The CORRECT order of bond energy is:

(A) $F_2 > Cl_2 > Br_2 > I_2$ (B) $Cl_2 < F_2 < Br_2 < I_2$

(C) $Cl_2 > Br_2 > F_2 > I_2$ (D) $I_2 > Br_2 > Cl_2 > F_2$

CPE217

60. Deacon's process of manufacture of chlorine is represented by the following equation?

(A) $MnO_2 + HCl \rightarrow MnCl_2 + Cl_2 + H_2O$

(B) $HCl + O_2 \xrightarrow[(723K)]{CuCl_2} Cl_2 + H_2O$

(C) $NaCl + MnO_2 + H_2SO_4 \rightarrow Cl_2 + MnCl_2 + NaHSO_4 + H_2O$

(D) $KMnO_4 + HCl \rightarrow KCl + MnCl_2 + H_2O + Cl_2$

CPE225

For More Join: @IITJEE_Advanced

61. An inorganic white crystalline compound (P) has a rock salt structure. (P) on reaction with conc. H_2SO_4 and MnO_2, evolves a pungent smelling, greenish-yellow gas (Q). Compound (P) gives white ppt. of (R) with $AgNO_3$ solution. Compounds (P), (Q) and (R) will be respectively:

(A) NaCl, Cl_2, AgCl (B) NaBr, Br_2, NaBr

(C) NaCl, Cl_2, Ag_2SO_4 (D) Na_2CO_3, CO_2, Ag_2CO_3

CPE254

Hydrochloric Acid(HCl): Preparation and Properties

62. HCl gas is dried by passing through:

(A) Conc. H_2SO_4 (B) Anhydrous $CaCl_2$ (C) P_2O_5 (D) Quick lime

CPE239

63. The INCORRECT statement about HCl is :

(A) It is a colourless, pungent smelling gas

(B) It is not combustible

(C) It changes blue litmus red, when dry

(D) It is extremely soluble in water.

CPE240

Oxoacids of Halogens: Preparation and Properties

64. Cl═O bond is NOT present in:

(A) Chloric acid (B) Hypochlorous acid

(C) Chlorous acid (D) Perchloric acid

CPE244

Rankers Academy JEE

65. The correct order of increasing oxidation number of Cl is :

(A) HOCl < HOClO < $HOClO_2$ < $HOClO_3$ (B) HOCl < $HOClO_2$ < HOClO < $HOClO_3$

(C) $HOClO_3$ < $HOClO_2$ < HOClO < HOCl (D) $HOClO_3$ < HOCl < $HOClO_2$ < HOClO

CPE245

Interhalogen Compounds

66. The hybridisation of I in cationic and anionic part of I_2Cl_6 are :

(A) sp^2, sp^3d (B) sp^3, sp^3 (C) sp^2, sp^3d^2 (D) sp^3, sp^3d^2

CPE252

Uses of Group - 17- Elements and Compounds

67. Salt producing family is :

(A) Nitrogen family (B) Carbon family (C) Halogen family (D) Inert gas family

CPE214

Group 18: Atomic and Physical Properties (Occurance, Electronic configuration, Atomic Radii, Ionisation Enthalpy, Electrongain Enthalpy)

68. The INCORRECT order is :

(A) He < Ne < Ar < Kr < Xe (Atomic Radius)

(B) He > Ne > Ar > Kr > Xe (Ionisation enthalpy)

(C) He < Ne < Ar < Kr < Xe (Density)

(D) He < Ne < Ar < Kr < Xe (Electron gain enthalpy)

CPE267

For More Join: @IITJEE_Advanced

69. The melting and boiling point of noble gases are low because : -

(A) Their size is very small

(B) They don't have free electrons

(C) The only type of interatomic interaction is weak dispersion forces (London force)

(D) They have no intermolecular attraction

CPE268

Chemical properties of Group 18 Elements

70. Which noble gas is not present in atmosphere?

(A) He (B) Ne (C) Xe (D) Rn

CPE264

71. The most abundant noble gas is :

(A) He (B) Ne (C) Ar (D) Kr

CPE265

72. Group 18 elements are called noble gas because :

(A) They react with a few elements only under certain conditions

(B) They don't react with any element at all

(C) They are similar to noble metals (Au, Ag, Pt)

(D) They are unstable

CPE266

Rankers Academy JEE

Compounds of Xenon with Oxygen and Fluorine, Chemical Properties

73. Which one of the following is not formed during hydrolysis of XeF_6 ?

(A) $XeOF_4$ (B) $XeOF_3$ (C) XeO_3 (D) XeO_2F_2

CPE274

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Group 13 Elements; The Boron Family (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

1. Which of the following are INCORRECT order of ionization enthalpy?

(A) B > Al > Ga > In > Tl (B) B > Tl > Ga > Al > In

(C) B > Al > Ga > Tl > In (D) B > Tl > Al > Ga > In

CPB125

Preparation of Boron Some Important compounds of Group-13 and their physical and chemical properties(Borax, Diborane, Orthoboric Acid, Alums)

2. Which of following are correct.

(A) B_2H_6 is non-planar.

(B) B_2H_6 undergo symmetrical cleavage with PF_3 , CO and $(C_2H_5)_3N$

(C) B_2H_6 undergo unsymmetrical cleavage with NH_3, CH_3NH_2, $(CH_3)_2NH$.

(D) $BeH_2(s)$, Ga_2Me_6 , Al_2Me_6 have two type of bonds (2C–2e bond as well as 3C–2e bond)

CPB195

Group 14 Elements; The Carbon Family (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

Rankers Academy JEE

3. Select the correct orders :

(A) C > Si > Ge > Sn > Pb (melting point) (B) C > Si > Ge > Pb > Sn (melting point)

(C) C > Si > Ge > Sn > Pb (Boiling point) (D) C < Si < Ge < Sn < Pb (Metallic character)

CPB151

Silicates

4. Select the correct statement (s)

(A) CH_3 $SiCl_3$ undergoes hydrolysis followed by inter molecular elimination of water to form a complex cross - linked polymer (i.e silicone)

(B) Silicone fluids are thermally stable.

(C) In sheet silicate, three oxygen atoms of each tetrahedral are shared with adjacent SiO_4^{4-} tetrahedrals.

(D) Silica is attacked by HF and NaOH.

CPB197

Zeolites

5. Which of the following is/are incorrect?

(A) Purest form of allotrope of carbon is diamond

(B) Zeolite is an example of silicate

(C) Graphite is less thermodynamically stable than diamond

(D) Total 32 polygons are present in C_{60} (Fullerene)

CPB152

For More Join: @IITJEE_Advanced

6. Which is/are incorrect about permutit or zeolite :

(A) By it both temporary and permanent hardness of water can not be removed.

(B) Its formula is $Na_2Al_2Si_3O_8.xH_2O$

(C) It softens hard water by exchanging its $Na^{\oplus}$ ions with $Ca^{2\oplus}$ and $Mg^{2\oplus}$ present in hard water.

(D) Exhausted zeolite can be regenerated by passing conc. solution of NaCl through it.

CPB174

7. Cation exchanger zeolite $Na_2Al_2Si_2O_8.xH_2O$ can exchange cation with which of the following in aqueous solution ?

(A) $Mg(HCO_3)_2$ (B) K_2SO_4 (C) $CaCl_2$ (D) Na_2CO_3

CPB175

Compounds of Lead

8. Which of the following statement(s) is /are true about compound 'P' and 'Q' in the following reaction:

$$Pb + \text{dil. } HNO_3 \xrightarrow{\Delta} P + Q\uparrow + H_2O$$

(A) Compound 'Q' is a paramagnetic colourless gas.

(B) Compound 'P' gives brown colour gas on heating.

(C) Compound 'Q' is also obtained by heating of copper with dil. HNO_3.

(D) Compound 'P' is PbO_2.

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CPB173

9. Which of following molecules acts as good oxidizing agent ?

(A) $NaBiO_3$ (B) PbO_2 (C) $TlCl_3$ (D) $PbCl_2$

CPB105

MATRIX MATCH TYPE

Compounds of Tin

10.

Column-I Compounds	**Column-II** Correct statement for compounds given
(P) $SnCl_2$	(1) Used in printing technology
(Q) Butter of tin	(2) Used for gilding purpose (in joining gold pieces)
(R) Mosaic gold	(3) Reducing agent
(S) Pink salt	(4) Mordant

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	3	2	1	(B)	3	4	2	1
(C)	2	1	3	4	(D)	1	3	4	2

CPB168

For More Join: @IITJEE_Advanced

Group 15 Elements (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

11. Select correct statement(s) :

(A) HF > HCl > HBr > HI (Thermal stability) (B) HF < HCl < HBr < HI (Reducing nature)

(C) HF > HCl >HBr > HI (Acidic character) (D) HF > HI > HBr > HCl (Boiling point)

CPE228

PCl_3 Preparation and Properties

12. When PCl_3 undergoes in hydrolysis then X oxyacid of phosphorus is formed then Select the correct statement about X :

(A) Two P – OH bond (B) One P – H bond

(C) One P = O bond (D) Three P – OH bond

CPE151

P_4O_{10} Preparation and Properties

13. Which of the following statement are correct for P_4O_{10} :

(A) No. of P – O – P Bond are 6 (B) No. of P – O Bond are 12

(C) No. of P = O Bond are 4 (D) No. of P – O Bond are (10)

CPE153

Simple and Mixed Oxides

Rankers Academy JEE

14. Which of the following statement (s) is/are correct?

(A) Lone pairs at central atom in H_2O and H_2S are present in pure p – orbital.

(B) H_2O is a better reducing agent than H_2Se

(C) H_2S is thermally more stable H_2Se.

(D) H_2Se is a weaker acid than H_2Te.

CPE177

Allotropes of Sulphur

15. Which of the following are incorrect?

(A) Both S_8 and S_6 molecules have crown shape structure.

(B) Magnetic property of S_2 molecule is same as $O_2^{2\ominus}$ ion.

(C) Rhombic sulphur is insoluble in water but soluble in CS_2.

(D) Both α and β sulphur have crown shape structure.

CPE208

Sulphuric Acid(H_2SO_4): Preparation and Properties

16. Which of the following salts will evolve halogen on treatment with conc. H_2SO_4?

(A) NaCl (B) KI (C) NaBr (D) None of these

CPE211

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Preparation of Boron Some Important compounds of Group-13 and their physical and chemical properties (Borax, Diborane, Orthoboric Acid, Alums)

1. Number of lone pair present in Al_2Cl_6.

CPB140

2. $B + HNO_3 \longrightarrow$ (A) + (B)

conc. white solid brown gas

How many of the following statements are correct regarding products (A) & (B)?

(1) (A) can be prepared by reacting borax with dilute acids.

(2) (A) is a weak monobasic lewis acid.

(3) (A) behaves like a strong electrolyte when dissolved in water.

(4) (A) can be prepared by reacting colemanite, SO_2 & H_2O.

(5) (B) is paramagnetic in nature.

(6) (B) is a mixed anhydride.

(7) (B) does not forms a dimer

(8) (B) reacts with NaOH(aq) undergoing disproportionation.

(9) Bond angle about central atom is greater in (A) than in (B). (Assume atom forming maximum number of bonds as central atom)

Bankers Academy JEE

CPB200

Chemical Properties of Group 14 Elements (Reactivity towards Oxygen, Water, Acids, Alkalies, Hydrogen, Halogen)

3. No of compounds producing gas on hydrolysis (with H_2O) is .....

(a) Al_4C_3 (b) BaC_2 (c) Mg_2C_3 (d) SiC

(e) B_2H_6 (f) Fe_3C

CPB232

Allotropes of Carbon

4. Consider a prototypical fullerene, C_{60}.

Let,

a = Number of 5-membered rings ;

b = Number of 6-membered rings

c = Number of π-bonds in C_{60}

Find the value of (3a – 2b + c)

CPB198

For More Join: @IITJEE_Advanced

Silicones

5. Total number of compound having 3-D network like structure.

Diamond, CO_2, SiO_2, SiC, graphite

CPB163

Silicates

6. Number of corner/oxygen atoms shared per tetrahedron in 3D silicates are 'x' and in pyroxene silicates are 'y' then value of $\frac{2(x+y)}{x}$ is :

CPB164

7. In cyclic silicates the number of corners shared per $SiO_4^{4\ominus}$ unit is :

CPB162

8. The silicate anion in the mineral kinoite is a chain of three SiO_4 tetrahedron that share corners with adjacent tetrahedra. The mineral also contains Ca^{2+} and Cu^{2+} and As^{2+} ion & water molecule in a 1 : 1 : 1 ratio. The formula of mineral is $Ca_2Cu_2Si_3O_n.\ 2H_2O$. Give value of 'n"

CPB202

Zeolites

9. Central atom may exhibit sp^3 hybridisation in how many of the following species :

(a) CO_2 (b) Graphite (c) Diamond

(d) CO (e) H_3BO_3 (aq) (f) Zeolites (Si-central)

(g) Silicones (Si) (h) Chlorosilane (Si) (i) Borax (Boron)

(j) Al_2Cl_6 (k) B_2H_6 (l) SiO_2 (solid)

(m) H_2CO_3 (n) $COCl_2$ (o) CH_4 (p) CCl_4

CPB199

Carbides

10. On hydrolysis of CaC_2, it gives a gas which on trimerization gives compound 'y'. What is the molar mass of Y?

CPB176

Compounds of Lead

11. If oxidation state of 'Pb' is 'x' in PbO_2 and oxidation state of 'O' is 'y' in H_2O_2 and oxidation state of 'N' is 'z' in $Pb(NO_3)_2$, then find out the value of 'z-x-|y|'.

CPB177

12. How many compounds show amphoteric nature amongst following

(a) B_2O_3 (b) Tl_2O_3 (c) $Al(OH)_3$ (d) $Ga(OH)_3$

(e) Al_2O_3 (f) Ga_2O_3 (g) $NaAlO_2$ (h) $Sr(OH)_2$

(i) Cr_2O_3

CPB201

Rankers Academy JEE

For More Join: @IITJEE_Advanced

Group 15 Elements (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

13. Sum of electronegativity value of Fluorine & Chlorine.

CPE238

Ammonia Preparation and Properties

14. Hydrolysis of NCl_3 give product x & y respectively than total number of σ bond in x + y.

CPE234

PCl_5 Preparation and Properties

15. Among the following the number of compounds that can react with PCl_5 to give $POCl_3$ is C_2H_5OH, CH_3COOH, SO_2, H_2O, H_2SO_4, P_4O_{10}

CPE156

Oxyacids of Phosphorous (preparation)

16. How many of the following is correct order of specified property.

(a)	N > P > As > Sb > Bi	(First ionisation enthalpy)
(b)	N > P > As > Sb	(Electronegativity)
(c)	N–N < P–P < As–As	(Single bond length)
(d)	$As^{3+} > Sb^{3+} > Bi^{3+}$	(Stability of +3 oxidation state)
(e)	White > Red > Black	(Reactivity of allotropes of phosphorus)
(f)	$H_3PO_2 < H_3PO_3 < H_3PO_4$	(Proticity of acids)
(g)	$H_3PO_2 < H_3PO_3 < H_3PO_4$	(Reducing power of acids)

CPE290

Rankers Academy JEE

P_4O_6 Preparation and Properties

17. The total number of lone pairs in P_4O_6 molecule is :

CPE155

P_4O_{10} Preparation and Properties

18. Total number of sp^3 hybridised atom in P_4O_{10} molecule.

CPE162

Group 16 Elements (Occurance, Electronic configuration, Atomic Radii and Ionic Radii. Ionisation Enthalpy, Electrongain Enthalpy, Electronegativity, Physical Properties)

19. Some statements are given below :

S_1 : Density of selenium is more than sulphur.

S_2 : $O^{\ominus}$ has greater ionization energy than $S^{\ominus}$.

S_3 : Oxygen has the highest electron affinity among the chalcogens.

S_4 : Electron gain enthalpy decreases down the group in group – 16 as the size increases.

If the number of correct statements are 'x' then find the value of $(x - 2)^2$.

CPE181

Dioxygen Preparation and Properties

20. From the given reaction :

$2KClO_3 \xrightarrow[MnO_2]{Heat} X + Y$

Bond order of gaseous molecule having paramagnetic nature.

CPE187

For More Join: @IITJEE_Advanced

Simple and Mixed Oxides

21. How many of the following oxides are acidic in nature?
SO_2, Cl_2O_7, Na_2O, CaO, Mn_2O_7, CrO_3, CO_2, N_2O_5, BaO

CPE180

Ozone Preparation and Properties

22. Some statements are given. If number of correct statements are P. and number of incorrect statements are Q then find the value of $\left(\frac{P}{Q}+P^2\right)$:

S_1 : Ozone is toxic but not more than KCN.
S_2 : Ozone has different colours in all three physical state.
S_3 : Turpentine oil can absorb ozone.
S_4 : Ozone layer protects earth from harmful ultraviolet rays coming from sun.
S_5 : Bond order of 'O–O' bond is not same in O_2 and O_3.

CPE182

Allotropes of Sulphur

23. Total number of non-bonding electron pairs in S_8 (crown shape) molecule.

CPE186

SO_2 Preparation and Properties

24. How many of the following statements are correct

S_1 : SO_2 reduces $Cr_2O_7^{2\ominus}$ to $Cr^{3\oplus}$.

S_2 : Yellow colour solution of $Fe^{3\oplus}$ changes to green colour in the presence of SO_2.
S_3 : SO_2 on reaction with water gives an acid having 2 π-bonds.
S_4 : solubility of H_2S in H_2O increases with decreasing temperatures.

CPE213

Rankers Academy JEE

Sulphuric Acid(H_2SO_4): Preparation and Properties

25. The sum of pπ – pπ and pπ – dπ bonds in H_2SO_4 molecule is :

CPE212

Hydrochloric Acid(HCl): Preparation and Properties

26. Complete hydrolysis product of 1 mole each of following will need how many total number of moles of NaOH for complete neutralisation?
$SOCl_2$, SO_2Cl_2 , PCl_3 , PCl_5 , NCl_3

CPE291

Oxoacides of Halogens: Preparation and Properties

27. Total number of pπ-pπ bonds in $HClO_4$.

CPE263

Interhalogen Compounds

28. How many of following interhalogen compounds do not exist ?
IF, ICl_3, $ClBr_3$, IBr_3, BrI_5, $BrCl_3$, FCl_3, ClF, IF_7

CPE260

29. Among the following:
ClF, ICl, IBr, ClF_3, IF_3, BrF_3, BrF_5, IF_7
'x' are gases and 'y' are solids. The sum x + y will be :

CPE261

Compounds of Xenon with Oxygen and Fluorine, Chemical Properties

30. XeF_4 reacts with PF_5 as follows : $XeF_4 + PF_5 \longrightarrow [XeF_3]^{\oplus}\ [PF_6]^{\ominus}$. If total numbers of 90° angles in reactants = x and if total number of 90° angles in products = y, considering all effects of VSEPR theory, then calculate the value of (y – x).

CPE288

For More Join: @IITJEE_Advanced

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

"Preparation of Boron Some Important compounds of Group-13 and their physical and chemical properties (Borax, Diborane, Orthoboric Acid, Alums)"

31. The total number of boron Atoms that have sp^3 hybridisation in one molecule of borax is.

CPB216

32. Number of B – O – B bonds in borax is x then value of $\frac{x}{10}$ is?

CPB217

33. Total number of atoms in a planar ring structure of H_3BO_3

CPB218

34. The number of water of crystallisation in borax are ___.

CPB219

35. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is

CPB220

Rankers Academy JEE

Uses of Boron and Aluminium and their compounds

36. The coordination number of Al in the crystalline state of $AlCl_3$ is

CPB221

Allotropes of Phosphorus; Phosphorus Preparation and Properties

37. Calculate the value of (x+y).z
Here
x = Atomicity of white phosphorus
y = Number of S–S bonds in one rhombic sulphur molecule
z = Atomicity of monoclinic sulphur

CPE330

Phosphine Preparation and Properties

38. Calculate the moles of phosphine formed when one mole of P_4 reacts with excess of NaOH.

CPE331

PCl_5 Preparation and Properties

39. Among the following, the number of compounds that can react with PCl_5 to give $POCl_3$ is
O_2, CO_2, SO_2, H_2O, H_2O, H_2SO_4, P_4O_{10}

CPE332

For More Join: @IITJEE_Advanced

Allotropes of Sulphur

40. At high temperature sulphur vapour is predominantly in the form of S_2 (gas) molecules. Assuming that the molecular orbitals for third row diatomic molecules (homonuclear) are analogous to those for second – row molecules. Answer the following questions –

P : The number of π bond(s) in S_2.
Q : The no. of unpaired electrons in S_2.
R : The bond order of S_2.
S : The bond order of S_2^{2-} disulphide ion formed by the gain of two electrons.

Find the value of expression : $\left|\frac{[(P+S)^3-(Q+R)^2]}{3}\right|$

CPE333

SO_2 Preparation and Properties

41. The number of reactions gives SO_2 gas :
(a) Cu + conc. $H_2SO_4 \rightarrow$
(b) Zn + dil. $H_2SO_4 \rightarrow$
(c) S + $H_2SO_4 \rightarrow$
(d) $FeS_2 + O_2 \rightarrow$
(e) $NaHSO_3$ + dil. $H_2SO_4 \rightarrow$
(f) $H_2SO_4 + PCl_5 \rightarrow$

CPE334

42. SO_2 reacts with chlorine gas (in presence of charcoal) to give Y. How many moles of NaOH will be consumed for complete neutralisation of an aqueous solution containing 2 moles of Y.

Rankers Academy JEE

CPE335

Sulphuric Acid(H_2SO_4): Preparation and Properties

43. Consider following reactions

$\underset{1/4\text{ mole}}{S_8(s)} + H_2SO_4(conc.) \rightarrow$ products

$\underset{3/8\text{ mole}}{S_8(s)} + HNO_3(conc.) \rightarrow$ products

How many moles of gaseous products are obtained?

CPE336

Interhalogen Compounds

44. In I_2Cl_6 molecule, the maximum number of atoms that may lie in the same plane is

CPE337

Compounds of Xenon with Oxygen and Fluorine, Chemical Properties

45. Xe react directly with F_2 in different condition (by volume mixture) of Xe & F_2 at suitable condition example :- $\underset{(2:1)}{Xe+F_2} \longrightarrow XeF_2$

Ratio of Xe and F_2 is given to make different Xenon $(Fluoride)_n$ products
$XeF_2 \Rightarrow Xe : F_2 = x : a = 1 : 0.5$
$XeF_4 \Rightarrow Xe : F_2 = x : b$
$XeF_6 \Rightarrow Xe : F_2 = x : c$

The value of a + b + c will be (up to two decimal places)

CPE338

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The correct statements from the following are:

(A) The decreasing order of atomic radii of group 13 elements is Tl > In > Ga > Al > B.

(B) Down the group 13 electronegativity decreases from top to bottom.

(C) Al dissolves in dil. HCl and liberate H_2 but conc. HNO_3 renders Al passive by forming a protective oxide layer on the surface.

(D) All elements of group 13 exhibits highly stable +1 oxidation state.

(E) Hybridisation of Al in $[Al(H_2O)_6]^{3+}$ ion is sp^3d^2.

Choose the **correct** answer from the options given below : **[JEE (Main) 2024]**

(1) (C) and (E) only
(2) (A), (C) and (E) only
(3) (A), (B), (C) and (E) only
(4) (A) and (C) only

CPB222

2. Given below are two statements :

Statement I : Boron is extremely hard indicating its high lattice energy

Statement II : Boron has highest melting and boiling point compared to its other group members.

In the light of the above statements, choose the most appropriate answer from the options given below

[JEE (Main) 2023]

(1) Statement I is incorrect but Statement II is correct
(2) Both Statement I and Statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both Statement I and Statement II is incorrect

Rankers Academy JEE

CPB214

3. Given below are two statements :

Statement I: Group 13 trivalent halides get easily hydrolyzed by water due to their covalent nature.

Statement II: $AlCl_3$ upon hydrolysis in acidified aqueous solution forms octahedral $\left[Al(H_2O)_6\right]^{3+}$ ion.

In the light of the above statements, choose the ***correct answer*** from the options given below :

[JEE (Main) 2024]

1. Statement I is true but statement II is false
2. Statement I is false but statement II is true
3. Both statement I and statement II are false
4. Both statement I and statement II are true

CPB224

4. Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "X" with 6-equivalents of "Y". "X" and "Y", respectively are : **[JEE (Main) 2022]**

(1) $B(OH)_3$ and NH_3
(2) B_2H_6 and NH_3
(3) B_2H_6 and HN_3
(4) NH_3 and B_2O_3

CPB207

For More Join: @IITJEE_Advanced

5. Identify the incorrect pair from the following: **[JEE (Main) 2024]**

(1) Fluorspar- BF_3 (2) Cryolite-Na_3AlF_6

(3) Fluoroapatite-$3Ca_3(PO_4)_2.CaF_2$ (4) Carnallite-$KCl.MgCl_2.6H_2O$

CPB225

6. Aluminium chloride in acidified aqueous solution forms an ion having geometry **[JEE (Main) 2024]**

(1) Octahedral (2) Square Planar

(3) Tetrahedral (4) Trigonal bipyramidal

CPB226

7. Boric acid in solid, whereas BF_3 is gas at room temperature because of **[JEE (Main) 2023]**

(1) Strong ionic bond in Boric acid

(2) Strong Vander Waal's interaction in Boric acid

(3) Strong hydrogen bond in Boric acid

(4) Strong covalent bond in BF_3

CPB211

8. The Lewis acid character of boron tri halides follows the order : **[JEE (Main) 2023]**

(1) $BBr_3 > BI_3 > BCI_3 > BF_3$ (2) $BCl_3 > BF_3 > BBr_3 > BI_3$

(3) $BF_3 > BCl_3 > BBr_3 > BI_3$ (4) $BI_3 > BBr_3 > BCl_3 > BF_3$

CPB212

9. The geometry around boron in the product 'B' formed from the following reaction is

$BF_3 + NaH \xrightarrow{450K} A + NaF$

$A + NMe_3 \rightarrow B$ **[JEE (Main) 2022]**

(1) trigonal planar (2) tetrahedral (3) pyramidal (4) square planar

CPB050

10. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Boric acid is a weak acid

Reason R : Boric acid is not able to release H^+ ion on its own. It receives OH^- ion from water and releases H^+ ion.

In the light of the above statements, choose the most appropriate answer from the options given below. **[JEE (Main) 2022]**

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

CPB208

11. Given below are two statements :

Statement I : The electronegativity of group 14 elements from Si to Pb gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the most appropriate from the options given below : **[JEE (Main) 2024]**

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

CPB227

Rankers Academy JEE

For More Join: @IITJEE_Advanced

12. Evaluate the following statements related to group 14 elements for their correctness.

(A) Covalent radius decreases down the group from C to Pb in a regular manner.

(B) Electronegativity decreases from C to Pb down the group gradually.

(C) Maximum covalence of C is 4 whereas other elements can expand their covalence due to presence of d orbitals.

(D) Heavier elements do not form pπ-pπ bonds.

(E) Carbon can exhibit negative oxidation states.

Choose the **correct** Answer from the options given below: **[JEE (Main) 2024]**

(1) (C), (D) and (E) Only (2) (A) and (B) Only

(3) (A), (B) and (C) Only (4) (C) and (D) Only

CPB229

13. The correct order of bond enthalpy (kJ mol^{-1}) is : **[JEE (Main) 2023]**

(1) Si – Si > C – C > Sn – Sn > Ge – Ge

(2) Si – Si > C – C > Ge – Ge > Sn – Sn

(3) C – C > Si – Si > Sn – Sn > Ge – Ge

(4) C – C > Si – Si > Ge – Ge > Sn – Sn

CPB213

14. The INCORRECT statement regarding the structure of C_{60} is : **[JEE (Main) 2021]**

(1) The six-membered rings are fused to both six and five-membered rings.

(2) Each carbon atom forms three sigma bonds.

(3) The five-membered rings are fused only to six-membered rings.

(4) It contains 12 six-membered rings and 24 five-membered rings.

Bankers Academy JEE

CPB204

15. Which pair of oxides is acidic in nature? **[JEE (Main) 2021]**

(1) B_2O_3, CaO (2) B_2O_3, SiO_2 (3) N_2O, BaO (4) CaO, SiO_2

CPB048

16. Match List-I with List-II **[JEE (Main) 2022]**

List-I (Si-Compounds)	**List-II (Si-Polymeric/other products)**
(A) $(CH_3)_4Si$	(I) Chain silicone
(B) $(CH_3)Si(OH)_3$	(II) Dimeric silicone
(C) $(CH_3)_2Si(OH)_2$	(III) Silane
(D) $(CH_3)_3Si(OH)$	(IV) 2D – Silicone

Choose the correct answer from the options given below:

(1) (A)–(III), (B)–(II), (C)–(I), (D)–(IV) (2) (A)–(IV), (B)–(I), (C)–(II), (D)–(III)

(3) (A)–(II), (B)–(I), (C)–(IV), (D)–(III) (4) (A)–(III), (B)–(IV), (C)–(I), (D)–(II)

CPB206

For More Join: @IITJEE_Advanced

17. Given below are two statements. **[JEE (Main) 2022]**

Statement I : Stannane is an example of a molecular hydride.

Statement II : Stannane is a planar molecule.

In the light of the above statement, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CPB209

18. Compound A used as a strong oxidizing agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is : **[JEE (Main) 2021]**

(1) PbO_2 (2) PbO (3) $PbSO_4$ (4) Pb_3O_4

CPB233

19. Choose the correct statements about the hydrides of group 15 elements.

A. The stability of the hydrides decreases in the order $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

B. The reducing ability of the hydrides increases in the order $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

C. Among the hydrides, NH_3 is strong reducing agent while BiH_3 is mild reducing agent.

D. The basicity of the hydrides increases in the order $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

Choose the most appropriate from the option given below: **[JEE (Main) 2024]**

(1) B and C only (2) C and D only

(3) A and B only (4) A and D only

CPE340

20. Given below are the two statements: one is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): There is a considerable increase in covalent radius from N to P. However from As to Bi only a small increase in covalent radius is observed.

Reason (R): covalent and ionic radii in a particular oxidation state increases down the group.

In the light of the above statement, choose the most appropriate answer from the options given below: **[JEE (Main) 2024]**

(1) (A) is false but (R) is true

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

CPE341

21. Which one of the following group-15 hydride is the strongest reducing agent? **[JEE (Main) 2021]**

(1) AsH_3 (2) BiH_3 (3) PH_3 (4) SbH_3

CPE300

Rankers Academy JEE

For More Join: @IITJEE_Advanced

22. A group 15 element, which is a metal and forms a hydride with strongest reducing power among group 15 hydrides. The element is : **[JEE (Main) 2021]**

(1) Sb (2) P (3) As (4) Bi

CPE295

23. Among the given oxides of nitrogen; N_2O, N_2O_3, N_2O_4 and N_2O_5, the number of compound/(s) having N–N bond is : **[JEE (Main) 2021]**

(1) 1 (2) 2 (3) 3 (4) 4

CPE310

24. Concentrated HNO_3 reacts with Iodine to give **[JEE (Main) 2022]**

(1) HI, NO_2 and H_2O (2) HIO_2, N_2O and H_2O

(3) HIO_3, NO_2 and H_2O (4) HIO_4, N_2O and H_2O

CPE312

25. The number of non-ionisable hydrogen atoms present in the final product obtained from the hydrolysis of PCl_5 is : **[JEE (Main) 2021]**

(1) 0 (2) 2 (3) 1 (4) 3

CPE302

26. Reaction of thionyl chloride with white phosphorus forms a compound [A], which on hydrolysis gives [B], a dibasic acid. [A] and [B] are respectively **[JEE (Main) 2023]**

(1) P_4O_6 and H_3PO_3 (2) PCl_3 and H_3PO_3

(3) PCl_5 and H_3PO_4 (4) $POCl_3$ and H_3PO_4

CPE050

27. Chemical nature of the nitrogen oxide compound obtained from a reaction of concentrated nitric acid and P_4O_{10} (in 4 : 1 ratio) is : **[JEE (Main) 2021]**

(1) acidic (2) basic (3) amphoteric (4) neutral

CPE299

28. Choose the correct statements from the following

A. All group 16 elements form oxides of general formula EO_2 and EO_3 where E = S, Se, Te and Po. Both the types of oxides are acidic in nature.

B. TeO_2 is an oxidising agent while SO_2 is reducing in nature.

C. The reducing property decreases from H_2S to H_2Te down the group.

D. The ozone molecule contains five lone pairs of electrons.

Choose the correct answer from the options given below: **[JEE (Main) 2024]**

(1) A and D only (2) B and C only (3) C and D only (4) A and B only

CPE343

29. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : H_2Te is more acidic than H_2S.

Reason R: Bond dissociation enthalpy of H_2Te is lower than H_2S.

In the light of the above statements. Choose the most appropriate from the options given below. **[JEE (Main) 2024]**

(1) Both A and R are true but R is NOT the correct explanation of A.

(2) Both A and R are true and R is the correct explanation of A.

(3) A is false but R is true.

(4) A is true but R is false.

CPE344

For More Join: @IITJEE_Advanced

30. The correct order of melting points of hydrides of group 16 elements is : **[JEE (Main) 2022]**

(1) $H_2S < H_2Se < H_2Te < H_2O$ (2) $H_2O < H_2S < H_2Se < H_2Te$

(3) $H_2S < H_2Te < H_2Se < H_2O$ (4) $H_2Se < H_2S < H_2Te < H_2O$

CPE307

31. Match List-I with List-II : **[JEE (Main) 2021]**

List-I **Industrial process**	**List-II** **Application**
(a) Haber's process	(i) HNO_3 synthesis
(b) Ostwald's process	(ii) Aluminium extraction
(c) Contact process	(iii) NH_3 synthesis
(d) Hall-Heroult process	(iv) H_2SO_4 synthesis

Choose the correct answer from the options given below :

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

CPE294

32. The strongest reducing agent among the following is: **[JEE (Main) 2024]**

(1) NH_3 (2) SbH_3 (3) BiH_3 (4) PH_3

CPE348

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33. Reduction potential of ions are given below: **[JEE (Main) 2024]**

ClO_4^-	IO_4^-	BrO_4^-
$E° = 1.19V$	$E° = 1.65V$	$E° = 1.74V$

The correct order of their oxidising power is:

(1) $ClO_4^- > IO_4^- > BrO_4^-$ (2) $BrO_4^- > IO_4^- > ClO_4^-$

(3) $BrO_4^- > ClO_4^- > IO_4^-$ (4) $IO_4^- > BrO_4^- > ClO_4^-$

CPE349

34. Which of the following compound CANNOT act as a Lewis base? **[JEE (Main) 2021]**

(1) NF_3 (2) PCl_5 (3) SF_4 (4) ClF_3

CPE296

35. Which one of the following correctly represents the order of stability of oxides, X_2O; (X = halogen) ? **[JEE (Main) 2021]**

(1) Br > Cl > I (2) Br > I > Cl (3) Cl > I > Br (4) I > Cl > Br

CPE304

36. The most stable trihalide of nitrogen is: **[JEE (Main) 2022]**

(1) NF_3 (2) NCl_3 (3) NBr_3 (4) NI_3

CPE305

For More Join: @IITJEE_Advanced

37. Given below are two statements:

Statement I: Chlorine can easily combine with oxygen to from oxides: and the product has a tendency to explode.

Statement II: Chemical reactivity of an element can be determined by its reaction with oxygen and halogens.

In the light of the above statements, choose the **correct** answer from the options given below.

[JEE (Main) 2023]

(1) Both the statements I and II are true

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

(4) Both the Statements I and II are false

CPE316

38. Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are :

[JEE (Main) 2021]

(1) 3, 1 and 1 (2) 4, 1 and 0 (3) 1, 1 and 3 (4) 1, 2 and 3

CPE301

39. The correct group of halide ions which can be oxidised by oxygen in acidic medium is

[JEE (Main) 2023]

(1) Br^- only (2) Cl^-, Br^- and I^- only

(3) Br^- and I^- only (4) I^- only

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CPE320

40. The difference between electron gain enthalpies will be maximum between: **[JEE (Main) 2023]**

(1) Ne and F (2) Ne and Cl (3) Ar and Cl (4) Ar and F

CPE317

41. XeF_4 reacts with SbF_5 to form

$[XeF_m]^{n+}[SbF_y]^{z-}$

m + n + y + z = _____ **[JEE (Main) 2023]**

CPE318

42. The difference in the oxidation state of Xe between the oxidised product of Xe formed on complete hydrolysis of XeF_4 and XeF_4 is _____ **[JEE (Main) 2023]**

CPE319

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The compound(s) which react(s) with NH_3 to give boron nitride (BN) is(are) :-

[JEE (Advanced) 2022]

(A) B (B) B_2H_6 (C) B_2O_3 (D) HBF_4

CPB090

2. Among the following, the correct statement(s) is/are :- **[JEE (Advanced) 2017]**

(A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure

(B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure

(C) BH_3 has the three-centre two-electron bonds in its dimeric structure

(D) The Lewis acidity of BCl_3 is greater than that of $AlCl_3$

CPB087

3. The crystalline form of borax has:- **[JEE (Advanced) 2016]**

(A) Tetranuclear $[B_4O_5(OH)_4]^{2\ominus}$ unit

(B) All boron atoms in the same plane

(C) Equal number of sp^2 and sp^3 hybridized boron atoms

(D) One terminal hydroxide per boron atom

CPB086

4. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is - **[JEE (Advanced) 2015]**

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CPB084

5. The correct statements(s) for orthoboric acid is/are :- **[JEE (Advanced) 2014]**

(A) It behaves as a weak acid in water due to self ionization

(B) Acidity of its aqueous solution increases upon addition of ethylene glycol

(C) It has a three dimensional structure due to hydrogen bonding.

(D) It is a weak electrolyte in water

CPB083

6. With respect to graphite and diamond, which of the statement(s) given below is (are) correct ?

[JEE (Advanced) 2014]

(A) Graphite is harder than diamond.

(B) Graphite has higher electrical conductivity than diamond.

(C) Graphite has higher thermal conductivity than diamond.

(D) Graphite has higher C–C bond order than diamond.

CPB082

7. The stoichiometric reaction of 516 g of dimethyldichlorosilane with water results in a tetrameric cyclic product **X** in 75% yield. The weight (in g) of **X** obtained is___.

[Use, molar mass (g mol^{-1}): H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

[JEE (Advanced) 2023]

CPB230

For More Join: @IITJEE_Advanced

8. Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are:- **[JEE (Advanced) 2015]**

(A) $(CH_3)SiCl_3$ and $Si(CH_3)_4$ (B) $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$

(C) $(CH_3)_2SiCl_2$ and CH_3SiCl_3 (D) $SiCl_4$ and $(CH_3)_3SiCl$

CPB085

9. A tin chloride **Q** undergoes the following reactions (not balanced) :- **[JEE (Advanced) 2019]**

$Q + Cl^{\ominus} \rightarrow X$

$Q + Me_3N \rightarrow Y$

$Q + CuCl_2 \rightarrow Z + CuCl$

X is a monoanion having pyramidal geometry. Both **Y** and **Z** are neutral compounds. Choose the correct option(s).

(A) The central atoms in **X** is sp^3 hybridized

(B) The oxidation state of the central atom in **Z** is +2

(C) The central atom in **Z** has one lone pair of electrons

(D) There is a coordinate bond in **Y**

CPB088

10. The reaction $Pb(NO_3)_2$ and NaCl in water produces a precipitate that dissolves upon the addition of HCl of appropriate concentration. The dissolution of the precipitate is due to the formation of **[JEE (Advanced) 2022]**

(A) $PbCl_2$ (B) $PbCl_4$ (C) $[PbCl_4]^{2\ominus}$ (D) $[PbCl_6]^{2\ominus}$

CPB091

Rankers Academy JEE

11. At room temperature, disproportionation of an aqueous solution of *in situ* generated nitrous acid (HNO_2) gives the species **[JEE (Advanced) 2024]**

(A) H_3O^+, NO_3^- and NO (B) H_3O^+, NO_3^- and NO_2

(C) H_3O^+, NO^- and NO_2 (D) H_3O^+, NO_3^- and N_2O

CPE351

12. The treatment of galena with HNO_3 produces a gas that is **[JEE (Advanced) 2022]**

(A) paramagnetic (B) bent in geometry

(C) an acidic oxide (D) colorless

CPE323

13. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is ___

(Given data : Molar mass of water = 18 g mol^{1-}) **[JEE (Advanced) 2019]**

CPE086

14. Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives a gas **Q**. The amount of $CuSO_4$ (in g) required to completely consume the gas **Q** is _____.

[Given : Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63]

[JEE (Advanced) 2022]

CPE352

For More Join: @IITJEE_Advanced

15. The species formed on fluorination of phosphorus pentachloride in a polar organic solvent are :

[JEE (Advanced) 2024]

(A) $[PF_4]^+$ $[PF_6]^-$ and $[PCl_4]^+$ $[PF_6]^-$
(B) $[PCl_4]^+$ $[PCl_4F_2]^-$ and $[PCl_4]^+$ $[PF_6]^-$
(C) PF_3 and PCl_3
(D) PF_5 and PCl_3

CPE353

16. Match the reactions (in the given stoichiometry of the reactants) in List-I with one of their products given in List-II and choose the correct option.

[JEE (Advanced) 2023]

	List-I		**List-II**
(P)	$P_2O_3 + 3H_2O \rightarrow$	(1)	$P(O)(OCH_3)Cl_2$
(Q)	$P_4 + 3NaOH + 3H_2O \rightarrow$	(2)	H_3PO_3
(R)	$PCl_5 + CH_3COOH \rightarrow$	(3)	PH_3
(S)	$H_3PO_2 + 2H_2O + 4AgNO_3 \rightarrow$	(4)	$POCl_3$
		(5)	H_3PO_4

(A) P → 2; Q → 3; R → 1; S → 5
(B) P → 3; Q → 5; R → 4; S → 2
(C) P → 5; Q → 2; R → 1; S → 3
(D) P → 2; Q → 3; R → 4; S → 5

CPE325

17. Among B_2H_6, $B_3N_3H_6$, N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is ––––––––

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[JEE (Advanced) 2019]

CPE088

18. Consider the following molecules : Br_3O_8, F_2O, $H_2S_4O_6$, $H_2S_5O_6$, and C_3O_2.

Count the number of atoms existing in their zero oxidation state in each molecule. Their sum is____.

[JEE (Advanced) 2023]

CPE326

Comprehension # 19 to 20

The reaction of Cl_2 gas with cold dilute and hot concentrated NaOH in water give sodium salt of two (different) oxoacids of chlorine P and Q respectively. The Cl_2 gas reacts with SO_2 gas, in presence of charcoal to give a product R. R reacts with white phosphorous to give a compound S. On hydrolysis, S gives as oxoacid of phosphorous T.

19. R, S and T, respectively are - **[JEE (Advanced) 2013]**

(A) SO_2Cl_2, PCl_5 and H_3PO_4
(B) SO_2Cl_2, PCl_3 and H_3PO_3
(C) $SOCl_2$, PCl_3 and H_3PO_2
(D) $SOCl_2$, PCl_3 and H_3PO_3

CPE081

20. P and Q, respectively, are the sodium salts of – **[JEE (Advanced) 2013]**

(A) Hypochlorus and chloric acid
(B) Hypochlorus and chlorus acid
(C) Chloric and perchloric acids
(D) Chloric and hypochlorus acids

CPE082

For More Join: @IITJEE_Advanced

21. The reaction of $HClO_3$ with HCl gives a paramagnetic gas, which upon reaction with O_3 produces

[JEE (Advanced) 2022]

(A) Cl_2O (B) ClO_2 (C) Cl_2O_6 (D) Cl_2O_7

CPE324

22. The incorrect statement(s) regarding, (i) HClO, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is(are)

[JEE (Advanced) 2015]

(A) The number of Cl = O bonds in (ii) and (iii) together is three

(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three

(C) The hybridization of Cl in (iv) is sp^3

(D) Amongst (i) to (iv), the strongest acid is (i)

CPE083

23. The reaction of Xe and O_2F_2 gives a Xe compound **P**. The number of moles of HF produced by the complete hydrolysis of 1 mol of **P** is _____.

[JEE (Advanced) 2022]

CPE354

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For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	C	C	C	D	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	A	B	B	C	D	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	A	C	C	A	D	B	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	C	C	B	B	A	D	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	C	D	A	C	B	D	C	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	A	B	B	A	C	B	C	C	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	A	A	C	B	A	D	C	D	C	D
Que.	71	72	73							
Ans.	C	A	B							

EXERCISE (O-2)

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Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	ABCD	BCD	ABCD	AC	AB	AC	ABC	ABC	B
Que.	11	12	13	14	15	16				
Ans.	ABD	ABC	ABC	CD	AB	BC				

EXERCISE (S)

1.	16	**2.**	7	**3.**	4	**4.**	26	**5.**	3
6.	3	**7.**	2	**8.**	10	**9.**	11	**10.**	78
11.	0	**12.**	5	**13.**	7	**14.**	5	**15.**	6
16.	5	**17.**	16	**18.**	10	**19.**	0	**20.**	2
21.	6	**22.**	20	**23.**	16	**24.**	3	**25.**	2
26.	24	**27.**	0	**28.**	3	**29.**	6	**30.**	2
31.	2.00	**32.**	0.20	**33.**	8.00	**34.**	8.00	**35.**	3.00
36.	4.00	**37.**	96.00	**38.**	1.00	**39.**	4.00	**40.**	2.67
41.	4.00	**42.**	8.00	**43.**	24.00	**44.**	8.00	**45**	25.50

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	2	1	1	3	4	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	4	4	2	4	3	1	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	3	1	2	1	4	2	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	3	2	2	4	1	1	4	4	2
Que.	41	42								
Ans.	11	2								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BC	ACD	ACD	6	BD	BD	222	B	AD	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	AD	288	2.38 or 2.39	B	D	4	6	A	A
Que.	21	22	23							
Ans.	C	D	4							

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